

Optimal Packing of Fifteen Equal Disks in a Circle: An Exact Computer-Assisted Certificate

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Abstract

We study the minimum radius of a circle containing fifteen pairwise interior-disjoint unit disks. We give an exact construction of radius

$$R_0 = 1 + \sqrt{1 + (2 + \cot(\pi/5))^2} = 4.521356964706164 \dots$$

and a computer-assisted candidate lower-bound certificate. A radial threshold reduces every putative packing with radius at most R_0 to four possible inner/outer counts. For each of the 760 dihedral classes of their cyclic binary words, an integer difference-constraint test either excludes the class or passes it to exact rational interval refinement. Refinement leaves one cyclic type, $(I, O, O)^5$, and confines its five free center radii to $[1.68, 1.72]$. An explicit local angular inequality then forces the symmetric candidate radius vector and excludes every smaller container. The certificate uses integer and rational arithmetic, including an explicit Taylor remainder bound for the trigonometric table. The repository contains a serialized subdivision certificate, a separate rational checker, and a one-command replay.

Verification status. The finite computations and the separate certificate checker were successfully replayed with the accompanying source bundle. This is a candidate computer-assisted proof, not an independently formalized result. The geometric reductions and the implementation have not yet received independent mathematical or peer review.

1 Introduction

Packing congruent disks in a circle is equivalent to maximizing the minimum distance among a fixed number of points in a disk. Graham proved the small cases through seven points, and Pirl treated the circle-packing cases through ten. Later computational work by Graham, Lubachevsky, Nurmela, and Ostergård tabulated best-known configurations and described the configurations for 12 through 20 disks as conjecturally optimal; their 15-disk entry is the familiar five-inner, ten-outer arrangement [1, 2, 3]. Ekanayake and LaFountain subsequently proved the 14-disk case and reported that the proved cases then known were $N \leq 14$ and $N = 19$ [4]. Thus the candidate addressed here is historically important, but its status must be stated carefully: the theorem below is supported by the explicit certificate and replay in this manuscript and still calls for independent checking.

Our proof strategy does not assume that a minimizing packing has fivefold symmetry. It first moves the outer centers to the container wall, then enumerates every cyclic inner/outer word and every compatible radial bin using necessary angular inequalities. The finite stage uses exact integer difference constraints and rational interval subdivision; only the final surviving cyclic type is

handled by a local analytic inequality. This combines a global finite reduction with a symmetry-free local argument. The finite certificate only rejects a radial box when a necessary condition fails; a surviving box is not treated as evidence that a packing exists.

2 Problem and main result

Let $p_1, \dots, p_{15} \in \mathbb{R}^2$ be the centers of fifteen unit disks. A circle of radius R centered at the origin contains the disks without interior overlap exactly when

$$\|p_i\| \leq R - 1 \quad (1 \leq i \leq 15), \quad \|p_i - p_j\| \geq 2 \quad (i \neq j). \quad (1)$$

Write R_{15}^* for the infimum of feasible R .

Theorem 2.1. *The minimum enclosing radius for fifteen pairwise interior-disjoint unit disks is*

$$R_{15}^* = R_0 = 1 + \sqrt{1 + (2 + \cot(\pi/5))^2}.$$

The upper bound comes from the familiar D_5 arrangement with five inner disks and ten outer disks, listed among the best-known configurations [3, 5]. The lower-bound strategy is independent of the tight-partition reduction used for fourteen disks [4].

3 An exact packing of radius R_0

Set

$$\varphi = \frac{\pi}{5}, \quad a_0 = \csc \varphi, \quad b_0 = \sqrt{1 + (2 + \cot \varphi)^2}, \quad \alpha = \varphi - \arcsin(1/b_0), \quad R_0 = 1 + b_0.$$

For $j = 0, \dots, 4$, take centers

$$I_j = (a_0, 2j\varphi), \quad O_j^- = (b_0, 2j\varphi - \alpha), \quad O_j^+ = (b_0, 2j\varphi + \alpha)$$

in polar coordinates. The five inner centers form a regular pentagon and the ten outer centers alternate around the circle [5].

Adjacent inner centers have distance $2a_0 \sin \varphi = 2$. The smaller of the two successive angular gaps between outer centers is $2(\varphi - \alpha)$, and

$$2b_0 \sin(\varphi - \alpha) = 2.$$

The other outer gap is larger because $\alpha > \varphi/2$. For an inner center and either of its two nearest outer centers, the cosine rule gives

$$a_0^2 + b_0^2 - 2a_0b_0 \cos \alpha = 4.$$

These are the closest pairs within each type; all other same-type or mixed pairs have a larger angular separation and hence a larger center distance. Every outer disk is tangent to the container boundary, and $a_0 < b_0$. Thus the arrangement is feasible in radius R_0 , proving $R_{15}^* \leq R_0$; [figure 1](#) shows the resulting placement.

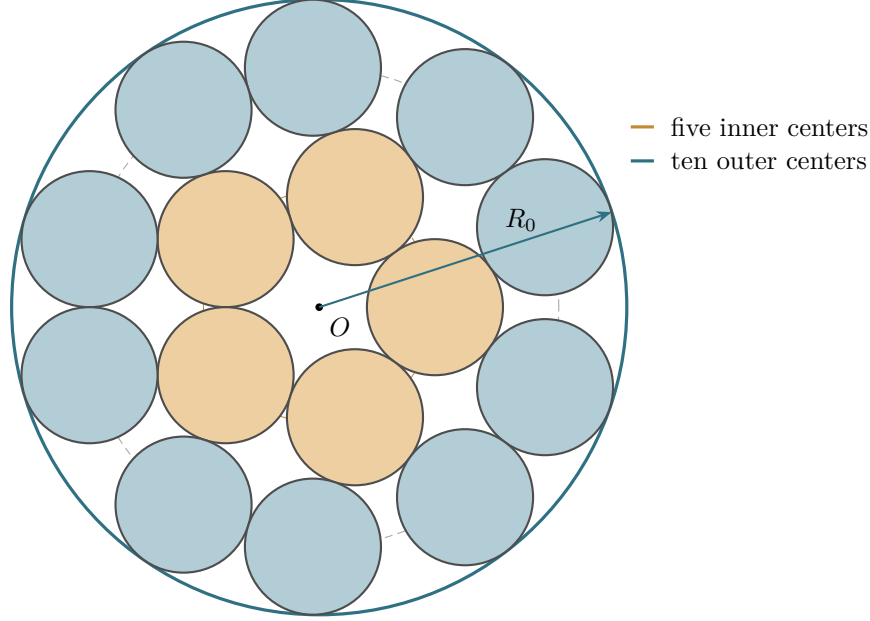


Figure 1: The exact fivefold-symmetric construction. The gold and blue circles represent unit disks centered at radii a_0 and b_0 ; the dashed circles mark those center radii. This verifies the upper bound only and makes no symmetry assumption about a minimizer.

4 Radial reduction

For a center at distance s from the origin, call it inner if $s < L$ and outer if $s \geq L$, where

$$L = b_0 - \frac{4}{b_0}. \quad (2)$$

The decimal enclosures used by the verifier are rational outward bounds; in particular $2.385431 < L < 2.385432$ and $b_0 < 3.522$.

For $x, y > 0$ define

$$c(x, y) = \frac{x^2 + y^2 - 4}{2xy}, \quad \Phi(x, y) = \arccos c(x, y)$$

whenever $|x - y| < 2 \leq x + y$. If the two center radii are x, y and their smaller angular separation is $\Delta \in [0, \pi]$, non-overlap requires $\Delta \geq \Phi(x, y)$.

Lemma 4.1 (Four-corner maximum). *On every positive rectangle $[u_1, u_2] \times [v_1, v_2]$, the maximum of $c(x, y)$ is attained at a corner.*

Proof. For fixed y ,

$$\frac{\partial c}{\partial x} = \frac{x^2 - y^2 + 4}{2x^2y}.$$

Its numerator is strictly increasing in x , so the derivative can change sign only from negative to positive. Thus an interior critical point in the x direction is a minimum, and a maximum occurs at an endpoint. Apply the same argument in y . $\square$

Lemma 4.2 (Pushing outer centers to the wall). *Suppose (1) holds with $R \leq R_0$. Each outer center can be moved radially to distance b_0 , with its polar angle fixed, without creating an overlap.*

Proof. An inner–outer pair has radii $x < L \leq y \leq b_0$. During the move, its squared distance has derivative

$$\frac{d}{dy}(x^2 + y^2 - 2xy \cos \Delta) = 2(y - x \cos \Delta) > 0,$$

so its distance cannot decrease. For an outer–outer pair, [lemma 4.1](#) and $L = b_0 - 4/b_0$ show that

$$\max_{L \leq x, y \leq b_0} c(x, y) = c(b_0, b_0),$$

with equality also at (L, b_0) and (b_0, L) . Its original angular separation therefore meets the required separation when both centers are moved to radius b_0 . $\square$

The two cases in this radial move are summarized in [figure 2](#).

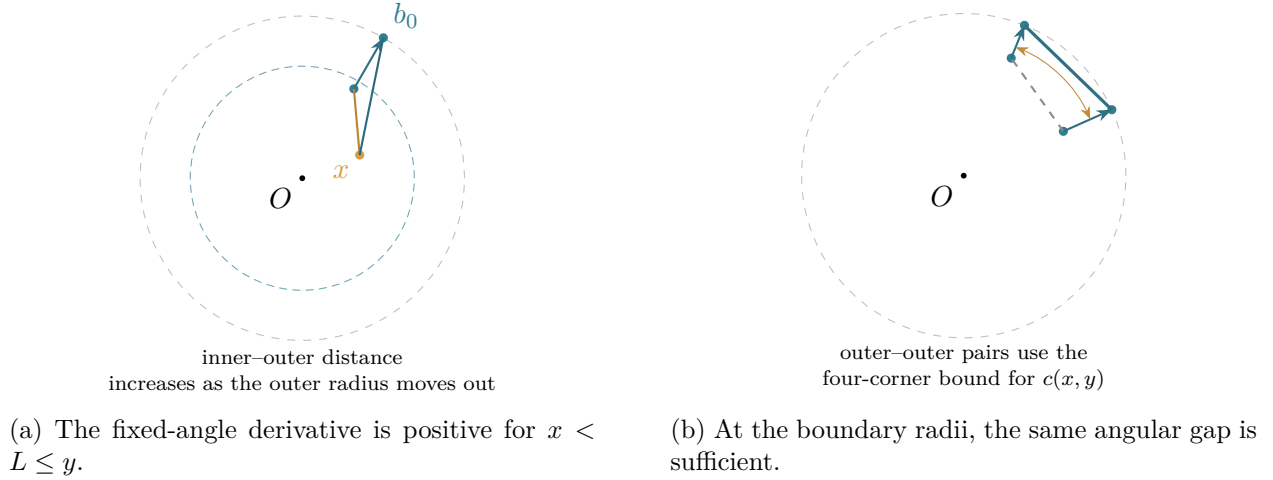


Figure 2: Schematic of the wall-pushing argument. The left panel is monotonicity of squared distance; the right panel summarizes [lemma 4.1](#). The dashed pair is illustrative, not a pointwise distance comparison throughout simultaneous motion.

Let $\theta_0 = 2 \arcsin(1/b_0)$. Any two outer centers, after the preceding move, require angular separation at least θ_0 . Since $11\theta_0 > 2\pi$, there are at most ten outer centers. Nine inner centers would be nine unit disks inside radius $L + 1 < 3.4$, whereas the exact nine-disk minimum radius is $1 + \sqrt{2(2 + \sqrt{2})} > 3.6$ [\[2, 5\]](#); hence there are at most eight inner centers. Since there are fifteen disks in all,

$$(|\mathcal{I}|, |\mathcal{O}|) \in \{(5, 10), (6, 9), (7, 8), (8, 7)\}. \quad (3)$$

The threshold and the surviving count ranges are sketched in [figure 3](#).

The standard small-packing bounds $s_{(5)} \geq a_0 > 5/3$ and $s_{(6)} \geq 2$ are also used in the radial interval enumeration. Here $s_{(j)}$ denotes the j th smallest center radius among all fifteen centers. They follow by applying the known exact five- and six-disk packing radii to the j innermost disks. Consequently, among k inner centers at least $k - 4$ have radius at least $5/3$, and at least $k - 5$ have radius at least 2.

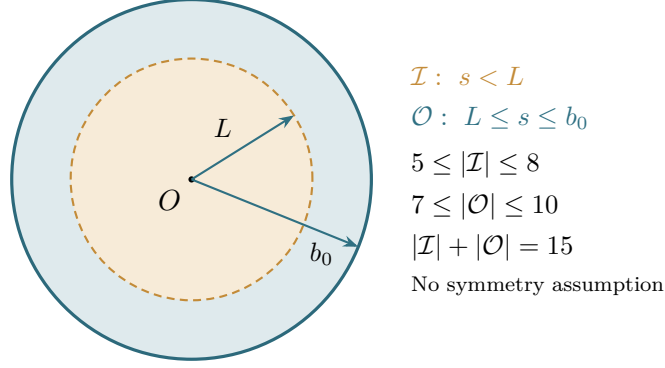


Figure 3: The radial threshold separates inner centers from the outer annulus. The count bounds leave only the four populations in [equation \(3\)](#); this is not a packing diagram.

5 Angular difference constraints

Order the centers by nondecreasing polar angle and lift their angles to $\theta_1 \leq \dots \leq \theta_{15} < \theta_1 + 2\pi$. For a pair whose radii lie in positive intervals X_i, X_j , let

$$C_{ij} = \max\{c(x, y) : x \in X_i, y \in X_j\}.$$

The maximum is obtained at the four interval corners by [lemma 4.1](#). The exact zero-radius case and positive intervals whose closure touches zero are handled separately; the endpoint rules and their bin ownership are proved in [section 6.1](#).

If $C_{ij} < 1$, put $\gamma_{ij} = \arccos C_{ij}$. Every feasible angle assignment must satisfy

$$\gamma_{ij} \leq \theta_j - \theta_i \leq 2\pi - \gamma_{ij} \quad (i < j). \quad (4)$$

If some box has $x + y < 2$ throughout, the box is impossible immediately. A box for which $|x - y| \geq 2$ throughout imposes no angular restriction for that pair. These rules are relaxed outwards, so every actual packing in a radius box satisfies the resulting difference constraints. The geometric conversion is illustrated in [figure 4](#).

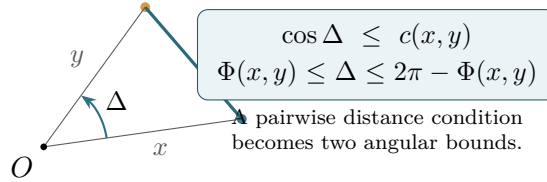


Figure 4: The law of cosines converts center separation into a lower bound on the smaller angular gap. The complementary upper bound in [equation \(4\)](#) is also needed for the difference-constraint graph.

Each inequality $\theta_v - \theta_u \leq w$ is a directed edge $u \rightarrow v$ with weight w . Such a system is feasible only if the directed graph has no negative cycle. The verifier replaces each required angle by a certified rational lower bound $q/2800$ and uses $44/7$ as an upper bound for 2π . Thus a lower-bound edge has weight $-q$, while its complementary upper-bound edge has weight $17600 - q$. A negative cycle in these integer weights excludes the entire radial box. The cancellation giving this contradiction is pictured in [figure 5](#).

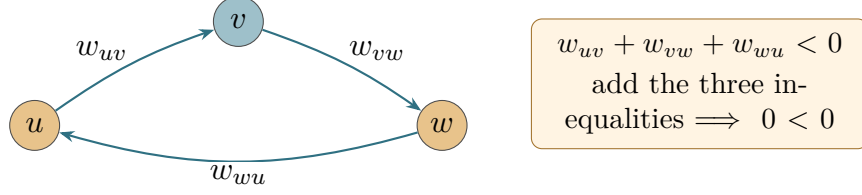


Figure 5: A negative directed cycle is a compact contradiction: summing its difference inequalities cancels all angle variables. The actual verifier uses exact integer edge weights for each radial box.

Certified trigonometric bounds. For $x \geq 0$, let $P_{18}(x) = \sum_{j=0}^9 (-1)^j x^{2j} / (2j)!$. Taylor’s theorem and $|\cos^{(19)}(t)| \leq 1$ give the rational lower bound

$$\cos x \geq P_{18}(x) - \frac{x^{19}}{19!}. \quad (5)$$

The explicit remainder subtraction is essential: the even Taylor polynomial alone is not used as a lower bound. For an interval pair with corner maximum C_{ij} , the integer q is chosen so that the right side of (5) at $x = q/2800$ is at least C_{ij} . Since cosine decreases on $[0, \pi]$, this certifies $q/2800 \leq \gamma_{ij}$. The integer range is certified by exact Machin-series bounds: $8790/2800 < \pi < 22/7$, so every tested angle is below π and $2\pi < 44/7$. All comparisons are performed with exact fractions.

6 Exhaustive finite certificate

The cyclic inner/outer word is defined up to rotation and reflection, because the disks are congruent and the container is rotationally and reflectively symmetric. Burnside’s lemma gives 111, 185, 232, and 232 classes for the four inner counts in (3), for a total of 760.

6.1 Half-open radial bins and the origin

The finite enumerators assign each radius to a half-open owner bin: every rational cutpoint belongs to the bin on its right. The first owner bin is split into the singleton $\{0\}$ and the positive interval $0 < r < 1/2$. The remaining owner bins begin at $1/2$, 1 , $3/2$, $8/5$, $5/3$, and 2 (with the stated further subdivisions in the twelve-bin stage); the last inner owner is truncated at L . The arrays used for interval bounds are closed rational hulls of these owner bins. Consequently, the outward bounds may include shared endpoints, but the discrete enumeration neither loses an endpoint nor relies on assigning it to two different cases. At an exact bisection, the two closed child hulls cover the parent, including their common midpoint.

This convention justifies the early count tests. The labels counted by ‘small’ have actual radii strictly below 1, so ‘small>1’ gives two centers with radial sum strictly less than 2. A first-bin center and a center in $[1, 3/2)$ also have sum strictly less than 2. The labels counted by ‘high’ and ‘high2’ have lower endpoints $5/3$ and 2 , respectively, so the rank bounds in the radial reduction are preserved at equality.

An exact-origin center has no polar angle. The graph therefore places no angular edges between it and other centers. Instead, since its distance to a center of radius s is exactly s , every other center must have radius at least 2. The refined enumerators enforce this directly: all other inner labels must be in bins with lower endpoint at least 2 (label 6 in the eight-bin grid and label 8 in the twelve-bin grid); the wall-pushed outer centers already have radius $b_0 > 2$.

For the positive first-bin type, $0 < x < 1/2$, the closed hull touches zero but does not represent the origin. When the other interval has upper endpoint at most 2, the cosine expression $c(x, y)$ is increasing in each variable on the relevant positive rectangle, since

$$\partial_x c = \frac{x^2 - y^2 + 4}{2x^2 y} \geq 0, \quad \partial_y c = \frac{y^2 - x^2 + 4}{2xy^2} \geq 0.$$

Thus the value at the upper endpoints is a valid upper bound even as $x \downarrow 0$. If the other interval extends above 2, the verifier disables that angular edge. The exact-origin singleton is never subjected to this limiting-angle rule.

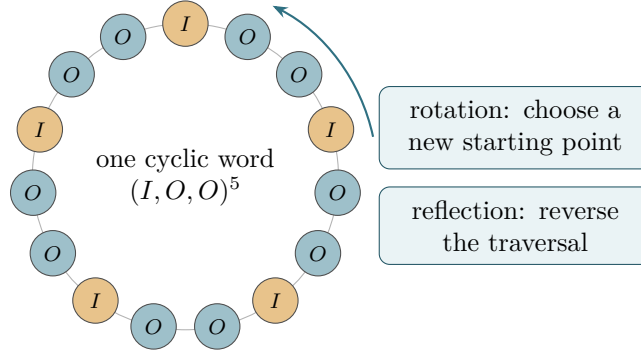


Figure 6: A binary bracelet records only the cyclic inner/outer order. Rotating or reversing the word changes its representation, not the underlying case; it does not make a geometric packing symmetric.

The finite verifier has four stages. First, it checks every positive entry in a fixed four-type angular table by exact rational arithmetic and applies its negative-cycle test to all 760 words. A word is not discarded by the separate analytic gap-capacity experiment. Second, the surviving words are refined over eight radial bins and then twelve radial bins, using the rank bounds in [section 4](#). Third, every remaining radial box is enumerated and split exactly until it is closed by a pair-sum contradiction, a negative cycle, or (for the canonical type only) containment in the local region. Fourth, a local analytic inequality closes the only remaining type and radius region. [figure 7](#) lays out these certified stages and their counts.

Table 1: Certificate reduction counts. A closed row means every represented radial box was excluded by a necessary condition.

Stage	Exact disposition	Count
Dihedral words	total	760
Coarse four-type graph	negative-cycle closures	382
Coarse four-type graph	passed to refinement	378
Eight-bin refinement	passed onward	23
Twelve-bin refinement	passed onward	4
Final types	$5I + 10O$ canonical, $5I + 10O$ noncanonical (2), $6I + 9O$ (1)	4

The exact rational subdivision closes the two noncanonical $5I + 10O$ words from 47 and 38 coarse radial boxes, respectively. Their branch-and-bound trees contain 1,641 and 646 nodes. The $6I + 9O$ word has three coarse boxes. Splitting its unique first-bin radius into $0 < r < 1/4$,

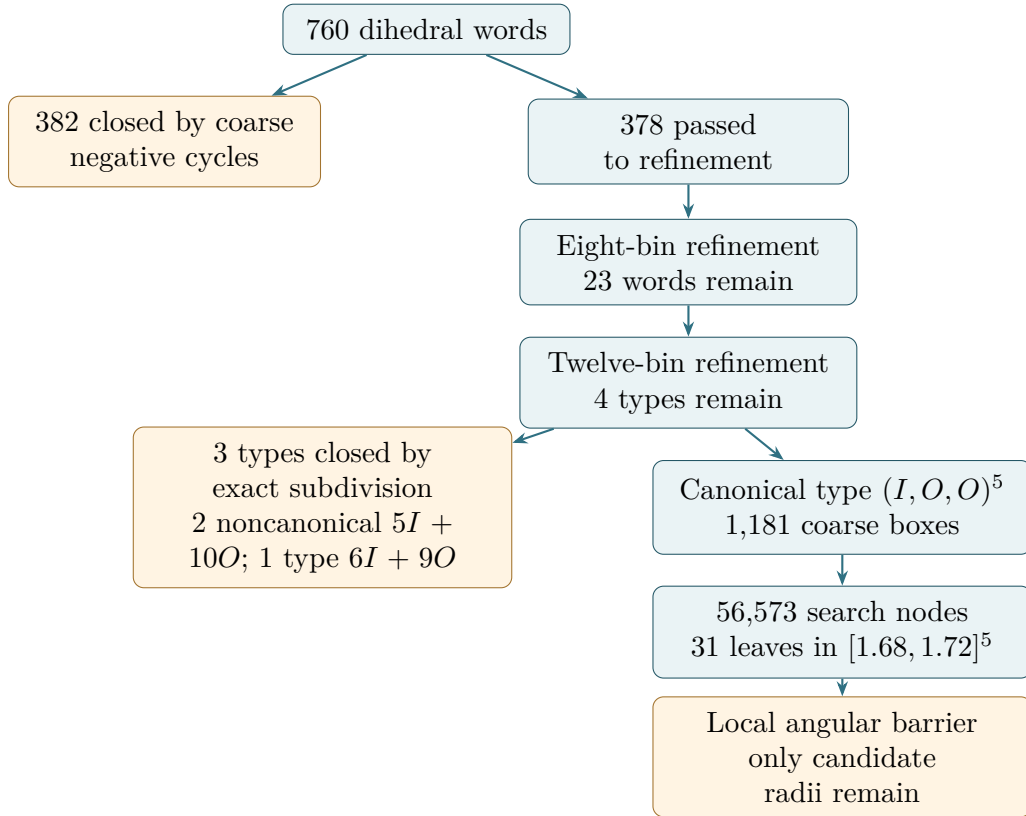


Figure 7: The replayed reduction path. Unlike an earlier exploratory gap-capacity filter, the certified path tests all 760 dihedral classes directly; the counts here match the included verifier reports.

$1/4 \leq r < 1/2$, and the exact origin gives nine leaves: six pair-sum contradictions and three negative cycles. Figure 8 shows the generic exact subdivision logic.

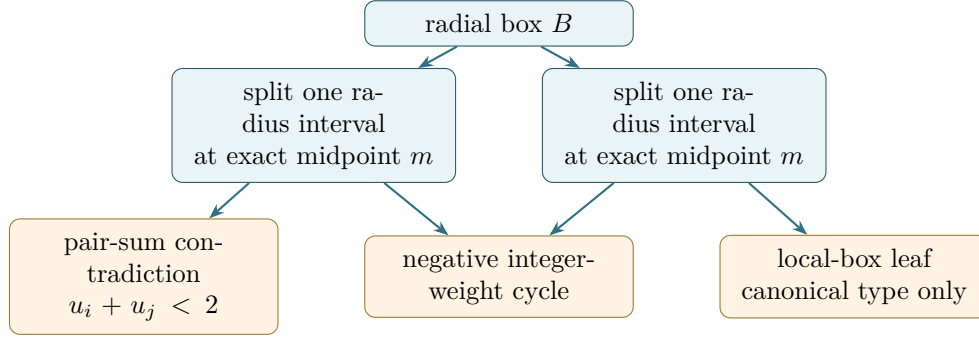


Figure 8: A schematic subdivision tree. Each split covers its parent by two closed rational children; a branch is finished only by a listed exact contradiction or, for the canonical type, by the local region proved separately below.

For the canonical word $(I, O, O)^5$, 1,181 coarse radial boxes generate 56,573 search nodes. There are 28,846 negative-cycle leaves, no pair-sum leaves, and 31 leaves contained in the local cube

$$[42/25, 43/25]^5 = [1.68, 1.72]^5.$$

The local argument below excludes every point of that cube except the candidate radius vector. It does not assume that the five radii or the packing are symmetric.

The complete data tables, branch rules, generated reports, and one-command replay are in the repository’s [candidate-bundle snapshot pinned to commit dbc19b12](#). The run script asserts the orbit counts, survivor identities, and finite-tree counts. It emits a 3.8 MB JSON certificate containing 1,266 exact subdivision roots, then `check_exact_certificate.py` independently verifies each bisection cover, pair-sum leaf, negative-cycle edge and weight, and local-region leaf using rational arithmetic. The checker also verifies the nine $6I + 9O$ endpoint leaves. It compares its roots with `full_residuals.txt`; completeness of the earlier coarse enumerations is established by exhaustive replay and count/set assertions, not by the serialized trees alone.

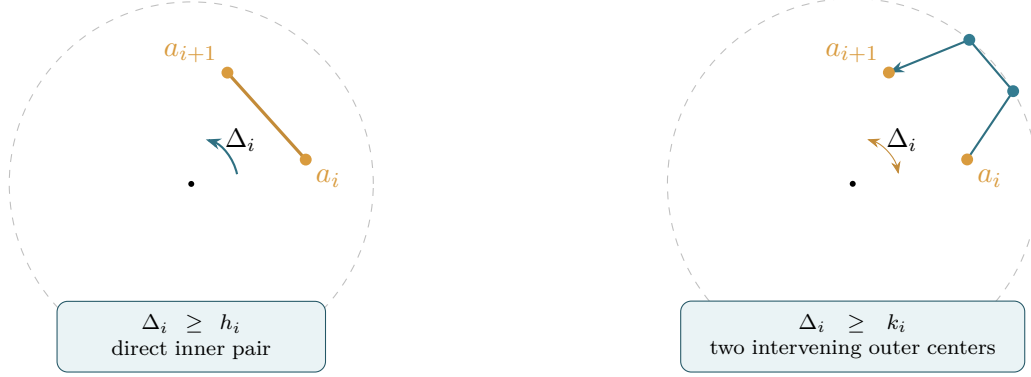
7 The local angular barrier

Assume the cyclic word is $(I, O, O)^5$. Let a_i be the five inner radii in cyclic order, with indices modulo five. Every outer radius is at most b_0 . Define

$$h_i = \Phi(a_i, a_{i+1}), \quad k_i = \Phi(a_i, b_0) + \Phi(b_0, b_0) + \Phi(a_{i+1}, b_0).$$

For $a_i \in [1.68, 1.72]$, the exact interval checks give $k_i > 1.175$. The four complementary inner-to-inner gaps therefore sum to more than $4.7 > \pi$, so the remaining gap is less than π . The direct inner pair and the path through the two intervening outer centers then give a lower bound $\max(h_i, k_i)$ on that gap. Summing the five gaps yields the necessary condition

$$\sum_{i=1}^5 \max(h_i, k_i) \leq 2\pi. \tag{6}$$



(a) The inner chord requires at least h_i .

(b) The three consecutive gaps sum to at least k_i .

Figure 9: Each inner-to-inner sector has two independent lower bounds. Combining them gives $\Delta_i \geq \max\{h_i, k_i\}$, which leads to the total angular budget in [equation \(6\)](#). The radii and angles are schematic.

When an outer center has radius below b_0 , the inner-outer angle requirement is at least the value used in k_i , because $\Phi(a, s)$ decreases with s on the relevant interval. The outer-outer term obeys the analogous bound.

At $a_i = a_0$, one has $h_i = k_i = 2\pi/5$. Write $\delta_i = a_i - a_0$ and $\sigma_i = \delta_i + \delta_{i+1}$. Differentiation at the candidate gives

$$\partial_x \Phi(x, y)|_{(a_0, a_0)} = -c_0, \quad \left. \frac{d}{dx} \Phi(x, b_0) \right|_{x=a_0} = d_0,$$

where

$$c_0 = \frac{\sin^2(\pi/5)}{\cos(\pi/5)} > \frac{21}{50}, \quad d_0 = \cos(\pi/5) > \frac{1}{2} > c_0.$$

The second derivatives on $[1.68, 1.72]^2$ and on $[1.68, 1.72] \times \{b_0\}$ are bounded, with exact rational checks, by

$$|\Phi_{xx}(x, y)| < 0.3, \quad |\Phi_{xy}(x, y)| < 0.8, \quad |\Phi_{xx}(x, b_0)| < 5.$$

Taylor's theorem therefore gives

$$\begin{aligned} h_i &\geq \frac{2\pi}{5} - c_0 \sigma_i - \frac{5}{2}(\delta_i^2 + \delta_{i+1}^2), \\ k_i &\geq \frac{2\pi}{5} + d_0 \sigma_i - \frac{5}{2}(\delta_i^2 + \delta_{i+1}^2). \end{aligned}$$

Since $d_0 > c_0 > 0$, $\max\{-c_0 \sigma_i, d_0 \sigma_i\} \geq c_0 |\sigma_i|$. The odd-cycle identity

$$2\delta_i = \sigma_i - \sigma_{i+1} + \sigma_{i+2} - \sigma_{i+3} + \sigma_{i+4}$$

implies $\sum_i |\sigma_i| \geq \frac{2}{5} \sum_i |\delta_i|$. Moreover $|\delta_i| \leq 11/500$ throughout the local cube. Hence

$$\begin{aligned} \sum_i \max(h_i, k_i) &\geq 2\pi + c_0 \sum_i |\sigma_i| - 5 \sum_i \delta_i^2 \\ &\geq 2\pi + \left(\frac{2}{5} c_0 - 5 \cdot \frac{11}{500} \right) \sum_i |\delta_i| \\ &> 2\pi + \frac{29}{500} \sum_i |\delta_i|. \end{aligned}$$

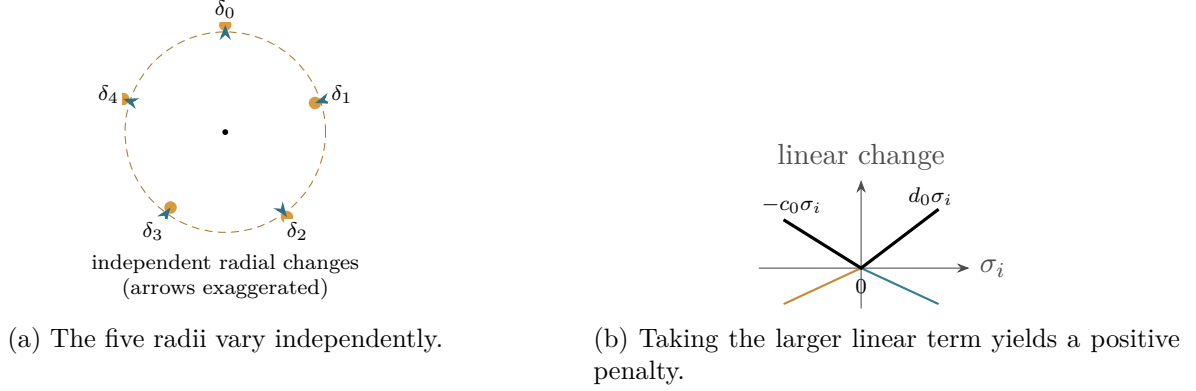


Figure 10: The local barrier has no hidden symmetry constraint: radial deviations are independent, while the maximum of the oppositely sloped first-order terms grows away from $\sigma_i = 0$. Quadratic errors are controlled by the exact interval bounds.

This contradicts (6) unless every $\delta_i = 0$.

If every $a_i = a_0$ but $R < R_0$, each original outer radius is strictly below b_0 . Its required inner–outer angle is then strictly larger than the candidate value, so the five-gap sum is strictly larger than 2π , again a contradiction. Thus no packing with $R < R_0$ exists in the final surviving class.

8 Conclusion and reproducibility

The construction in section 3 proves $R_{15}^* \leq R_0$. The radial classification, exhaustive difference-constraint certificate, exact rational subdivision, and local angular barrier prove that every feasible packing has $R \geq R_0$. This proves theorem 2.1, subject to the correctness of the finite certificate and its stated geometric reductions.

To replay the computations, use Python 3.10 or newer, GNU C++17, and a POSIX shell:

```
cd research/fifteen_equal_completion/candidate_bundle
bash run_all.sh
```

All exclusion decisions use integers or rational fractions. The code’s decimal output is diagnostic only. The replay is self-contained and records the branch counts and terminal reasons; independent reimplementations and peer review remain desirable for a result of this scope.

References

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- [2] U. Pirl, “Der Mindestabstand von n in der Einheitskreisscheibe gelegenen Punkten,” *Mathematische Nachrichten* 40 (1969), 111–124.
- [3] R. L. Graham, B. D. Lubachevsky, K. J. Nurmela, and P. R. J. Ostergård, “Dense packings of congruent circles in a circle,” *Discrete Mathematics* 181 (1998), 139–154. [https://doi.org/10.1016/S0012-365X\(97\)00050-2](https://doi.org/10.1016/S0012-365X(97)00050-2).

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